Frequency Hopping and BER Primer: Frequency-hopping spread spectrum.

In frequency-hopping spread spectrum (FHSS) the transmitter rapidly switches its carrier among many frequency channels following a pseudo-random sequence that the receiver also knows. The time spent on one frequency is the dwell time, and the number of changes per second is the hop rate.

FHSS resists narrowband interference, because only the hops that land on the interfered channel are damaged. It also makes eavesdropping harder and averages out frequency-selective fading. Slow hopping sends several symbols per hop, while fast hopping uses several hops per symbol.

Classic Bluetooth is a well-known FHSS system. It hops over 79 channels of 1 MHz in the 2.4 GHz ISM band at 1600 hops per second.

Frequency Hopping and BER Primer: Modulation and bit error rate.

For coherent BPSK over an additive white Gaussian noise (AWGN) channel the bit error rate is Q(sqrt(2 Eb/N0)), which equals 0.5 times erfc(sqrt(Eb/N0)). Here Eb is the energy per bit and N0 is the noise power spectral density.

QPSK with Gray coding has the same bit error rate as BPSK for a given Eb/N0, while sending two bits per symbol and so using half the bandwidth. BPSK needs an Eb/N0 of about 9.6 dB to reach a bit error rate of 1e-5, and about 6.8 dB for 1e-3.

Simulating a link means generating random bits, mapping them to symbols, adding Gaussian noise scaled to the target Eb/N0, demodulating and counting errors. To estimate a very low error rate with reasonable confidence, enough error events must be observed, typically at least a hundred.

Frequency Hopping and BER Primer: Synchronisation and partial-band interference.

A frequency-hopping receiver must align both the timing and the hop sequence with the transmitter. Acquisition searches for the correct hop phase, and tracking then keeps the receiver locked as clocks drift.

In partial-band interference an interferer covers only a fraction of the hopping band. For a worst-case fraction, the bit error rate of uncoded BPSK falls only inversely with the signal-to-interference ratio, instead of exponentially as in plain AWGN. This is why FHSS systems rely on error-correcting codes, interleaving and diversity to survive intentional or accidental jamming.